

BERN BELPMOOS AP, Switzerland

WMO: 066300

Lat: 46.906N

Lon: 7.505E

Elev: 1673

StdP: 13.83

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.6	18.0	6.9	8.3	13.6	15.4	12.8	20.2	17.6	47.0	15.1	45.3	4.4	50	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	89.2	69.8	85.7	68.4	82.1	66.6	70.9	84.8	69.1	82.1	67.7	79.2	5.9	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	103.3	76.2	64.7	97.3	73.5	63.0	91.7	71.6	35.7	85.3	34.2	82.3	33.0	79.6	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.7	13.0	10.9	DB	6.8	94.0	6.4	2.9	2.2	96.1	-1.5	97.7	-5.1	99.3	-9.7	101.4	(4)
(5)			WB	6.3	74.3	6.4	1.5	1.7	75.4	-2.1	76.3	-5.7	77.1	-10.4	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	34.1	34.3	42.3	49.5	55.7	64.2	67.1	66.7	59.2	51.2	41.7	34.6		
	DBStd	13.63	7.00	7.71	5.86	6.31	6.28	5.94	5.60	5.62	5.59	6.19	6.06	6.85		
	HDD50	2085	493	441	245	84	19	0	0	0	2	65	257	479		
	HDD65	5715	957	861	705	464	294	86	40	49	188	428	700	943		
	CDD50	2131	1	0	5	70	196	425	530	518	278	101	7	0		
	CDD65	288	0	0	0	0	5	61	106	102	14	0	0	0		
	CDH74	3461	0	0	0	22	126	807	1244	1093	168	1	0	0		
	CDH80	1134	0	0	0	1	14	274	450	377	18	0	0	0		
(14) Wind	WSAvg	4.1	4.3	4.0	4.3	4.5	4.7	4.5	4.5	4.1	3.6	3.5	3.6	3.9		
(15) Precipitation	PrecAvg	50.3	2.7	2.9	3.2	3.9	5.6	5.7	5.8	5.6	4.3	3.6	3.5	3.4		
	PrecMax	63.0	6.4	6.9	9.8	7.4	9.1	10.2	13.3	9.8	6.7	6.3	7.7	6.9		
	PrecMin	39.2	0.8	0.9	1.5	1.0	1.9	2.5	1.8	1.0	2.2	0.5	0.2	0.3		
	PrecStd	6.3	1.5	1.6	1.7	1.7	2.2	1.8	2.3	2.1	1.5	1.4	1.8	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	59.4	67.9	77.3	82.2	91.6	93.5	91.7	82.4	73.4	63.0	54.0		
		MCWB	50.2	48.1	52.5	59.1	63.4	72.4	71.2	69.7	65.8	59.4	54.4	48.0		
	2%	DB	50.4	55.0	64.3	72.0	77.3	87.6	89.4	87.8	78.9	69.6	59.1	51.7		
		MCWB	45.1	46.1	50.5	56.3	61.1	69.3	68.9	69.4	65.1	59.0	52.7	45.8		
	5%	DB	46.8	49.7	60.6	69.6	73.7	82.6	85.6	84.4	75.2	64.8	54.0	48.0		
		MCWB	43.0	43.4	49.0	54.9	60.2	67.3	67.7	68.2	63.1	57.4	49.3	43.9		
	10%	DB	43.2	44.8	55.6	64.6	70.0	78.6	80.8	80.5	71.4	61.2	50.3	44.5		
		MCWB	40.6	40.6	46.8	52.9	58.1	65.3	65.8	67.0	61.5	55.8	46.7	41.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.1	49.8	53.7	59.7	65.7	74.1	72.5	72.4	67.5	62.0	55.3	49.4		
		MCDB	55.7	57.3	64.7	74.2	77.8	90.0	88.7	85.7	78.7	69.2	62.3	53.5		
	2%	WB	46.7	46.0	50.9	57.7	63.4	70.6	70.1	70.7	65.8	59.9	52.8	46.8		
		MCDB	50.7	52.4	61.8	71.3	73.9	84.0	84.8	84.1	76.3	66.1	58.2	50.7		
	5%	WB	43.2	43.4	49.1	55.5	61.1	68.3	68.5	69.0	64.2	58.2	50.2	44.3		
		MCDB	46.1	48.6	58.7	67.1	71.3	80.3	81.3	81.7	73.0	64.0	54.2	47.5		
	10%	WB	40.8	40.8	47.5	53.6	59.0	66.0	67.1	67.6	62.5	56.6	47.8	41.3		
		MCDB	42.8	44.4	55.7	63.5	68.3	76.3	78.5	78.4	69.5	61.4	49.7	43.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	16.2	21.4	22.8	21.2	22.1	22.9	23.2	21.2	18.5	13.8	13.5		
		MCDBR	17.1	25.8	32.0	33.1	29.5	29.9	30.6	29.6	26.7	24.9	21.1	17.7		
		MCWBR	12.6	17.1	19.3	17.7	14.9	13.8	12.3	12.7	13.8	15.7	14.6	12.9		
	5% WB	MCDBR	16.2	24.0	28.4	29.7	25.7	28.2	27.5	27.6	24.6	22.0	20.3	16.6		
		MCWBR	12.2	16.1	17.4	16.5	13.5	13.5	12.3	12.7	13.1	14.3	14.1	12.6		
(40) Clear-Sky Solar Irradiance	taub		0.290	0.317	0.358	0.398	0.412	0.414	0.414	0.403	0.379	0.353	0.317	0.288		
	taud		2.415	2.342	2.283	2.216	2.230	2.272	2.268	2.309	2.358	2.412	2.449	2.430		
	Ebn at Noon		255	269	272	271	271	269	267	263	252	242	242	242		
	Edh at Noon		22	29	36	42	43	41	41	38	33	27	21	19		
(44) All-Sky Solar Radiation	RadAvg		398	669	1039	1383	1589	1882	1831	1594	1234	790	434	323		
	RadStd		48	100	154	201	145	176	160	147	112	78	56	56		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (114 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+1.35	+0.74	+0.56	N/A	N/A	+103	+29

Nomenclature: See separate page